

Lab 2 Software Installation Manual

Keil uVision IDE Software Installation:

Version: 5.39.0.0 (Fill up following form to get the Download Link)

Download Link: <https://www.keil.com/demo/eval/arm.htm>

Video Link: <https://www.youtube.com/watch?v=buQtJjgw2pE>

Lab 1 Software Installation Manual

Logisim Evolution Installation:

Install version 3.8.0 using link: [Link](#)

Logisim-evolution is a Java application; therefore, it can run on any operating system supporting the Java runtime environment. It requires Java 16 (or newer). If JAVA is not installed, install using link: [Link](#)

Do not install Logisim Software which is different from Logisim Evolution Software. We will be using Logisim Evolution Software.

Those who have Intel Laptop with Win OS should install following Logisim Evolution File: logisim-evolution-3.8.0-x86.msi

Verilog Installation:

To download and install Iverilog, Goto this site: [Link](#)

Follow this YouTube tutorial: [Link](#)

Make sure to click "Add Icarus Verilog executables folder to the system PATH"

For Linux installation,
`sudo apt-get update`
`sudo apt-get install iverilog`
`sudo apt-get install gtkwave`

VSCoDe (IDE to write Verilog code):

My suggestion is VSCoDe. Goto to this link to download: [Link](#)

Follow this YouTube tutorial: [Link](#)

Make sure to install the Verilog addon for syntax highlighting.

Simulation of Verilog program with test bench:

Go to terminal in VSCode and write:

```
iverilog -o out.vvp comparator.v stimulus.v  
# comment: out.vvp is the name output file after  
# compilation. "-o" defines output parameter.  
vvp out.vvp  
# comment: Show the output of the stimulus
```

Graphical simulation/waveforms of Verilog program with test bench:

Go to terminal in VSCode and write:

```
gtkwave test.vcd &  
# It will show graphical simulation.
```

Slides link for Lab 1 Class:

[Introduction 3 - VLSI & FPGA v2.pdf](#)

[Introduction 2 - Analog Electronics II \(CMOS\) v2.pdf](#)

[ALU 1 - Adder & Subtractor v2.pdf](#)

[HDL 1 - Combinational Circuit.pdf](#)

[HDL 2 - Sequential Circuit.pdf](#)

Slides link for Lab 2 Class:

[uVision Project Tutorial](#) (STM32F103C8 should be selected instead of STM32F100RB) (To install STM32F103C8 in Pack Installer, write STM32F103C8 in search, then install Keil::STM32F1xx_DFP and Keil::STM32NUCLEO_BSP in Packs)

[ARM Assembly Slide](#)

Reading: Book 1 Chapter 6